

Final pair identification: closing the support-size 2^a theorem for $\lambda = (2^a, 1^{n-2a})$

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Abstract

We close Conjecture 5.6 of `2026-05-17-trajectory-uniform.tex` (“**May-17**”), the final residue of the support-size 2^a program for the family $\lambda = (2^a, 1^{n-2a})$ at $n \geq 3a$. The proof tracks the column-2 entry set $\sigma(P_k^+)$ through the trajectory and identifies it via a closed-form description of $S_a^{(n)}$:

$$T_\varepsilon = \{i-1 : \varepsilon_i = L, 1 \leq i \leq a\} \cup \{2a-i : \varepsilon_i = R, 1 \leq i \leq a\} \quad (\text{shifted by } -(n-2a+1)).$$

The trajectory’s evolution then breaks into three independent pieces: the lower range (controlled by the May-17 σ -invariant), an “upper” range (controlled by a one-line recursion on τ), and a “middle” pair $\{a-1, a\}$ in the shifted universe (controlled by the case at the final step). Matching the three pieces against the closed form yields the final pair identification

$$\{\bar{\sigma}(P_{a-1}^+(T')), \bar{\sigma}(P_{a-1}^-(T'))\} = \{T_{\tau L}, T_{\tau R}\}, \quad \tau := \varepsilon(T') \in \{L, R\}^{a-1}.$$

Consequence. Combined with the May-17 dimension bound $\dim B_{n-a+1}^{(\lambda)} V_\lambda \leq 1$ (May-17 §4.2) and the generic non-vanishing argument, we obtain **Conjecture 5.6 of May-17 unconditionally**: $B_{n-a+1}^{(\lambda)} V_\lambda$ is exactly 1-dimensional with seminormal support $S_a^{(n)}$, $|S_a^{(n)}| = 2^a$, all coefficients nonzero in $\mathbb{Q}(q)$. The rank-zero theorem for the family $\lambda = (2^a, 1^{n-2a})$ at every $a \geq 1$, $n \geq 3a$ holds (modulo Phase A endpoint).

1 Setup and notation

We adopt the conventions of May-17. Fix $a \geq 2$, $n \geq 3a$, $\lambda = (2^a, 1^{n-2a}) \vdash n$, $\mu_1 = (2^{a-1}, 1^{n-2a}) \vdash n-2$.

Encoding. For $\sigma \in \binom{[2a]}{a}$ in the recursive set S_a (PROVE.md / May-14), the encoding $\varepsilon : S_a \rightarrow \{L, R\}^a$ is defined recursively: $\varepsilon_1 = L$ if $0 \in \sigma$, R if $2a-1 \in \sigma$; strip that boundary element; shift the remainder down by 1; recurse on $S_{a-1} \subseteq [2a-2]$ to obtain $\varepsilon_2, \dots, \varepsilon_a$. For $T \in S_a^{(n)}$ (an SYT of λ in the support, i.e. in the S_a -indexed support stable at $n \geq 3a$), write $\bar{\sigma}(T) := \sigma(T) - (n-2a+1) \subseteq [2a]$ (the column-2 entries shifted into the universe $\{0, 1, \dots, 2a-1\}$).

For $T' \in S_{a-1}^{(n-2)}$, $\tau := \varepsilon(T') \in \{L, R\}^{a-1}$.

Trajectory. As in May-17, the trajectory is

$$w_k(T') = (T_{n-1-k}+1) \cdots (T_{n-2}+1) \Phi_1(v_{T'}), \quad k = 0, 1, \dots, a-1,$$

with $w_0 = \Phi_1(v_{T'})$. By Theorem 2.1 of May-17 (per-trajectory invariant), w_k has seminormal support $\{P_k^+, P_k^-\}$, a non-degenerate T_{n-1-k} -2-block. We use the May-17 convention $P_0^+ := T_A(T')$ (the extension with $n-1$ at $(a, 2)$, n at $(n-a, 1)$), and $P_{k+1}^+ :=$ the survivor of $\{P_k^+, P_k^-\}$ under $(T_{n-2-k+1})$ for $k \geq 0$.

We refer freely to the four-case structural Lemma 3.1 of May-17 (*Kill/expand alternation*) with the case-labelling I, I.a, I.b, II, II.a, II.b therein.

2 Closed form for S_a

Lemma 1 (Closed form). *For every $a \geq 1$ and every $\varepsilon \in \{L, R\}^a$,*

$$T_\varepsilon = \{i-1 : \varepsilon_i = L, 1 \leq i \leq a\} \cup \{2a-i : \varepsilon_i = R, 1 \leq i \leq a\}.$$

In particular, $T_\varepsilon \cap \{0, 1, \dots, a-1\} = \{i-1 : \varepsilon_i = L\}$ and $T_\varepsilon \cap \{a, \dots, 2a-1\} = \{2a-i : \varepsilon_i = R\}$ (so the L - and R -bits “fill” disjoint halves of $[2a]$).

Proof. Induction on a .

Base $a = 1$. $S_1 = \{\{0\}, \{1\}\}$. $\varepsilon = L$ gives $\{0\} = \{1-1\}$ (one L -contribution at $i = 1$); $\varepsilon = R$ gives $\{1\} = \{2-1\}$ (one R -contribution at $i = 1$). ✓

Inductive step $a-1 \rightarrow a$. Let $\varepsilon = (\varepsilon_1, \dots, \varepsilon_a)$. By the S_a recursion of PROVE.md / May-14, $T_\varepsilon = \{0\} \cup \text{shift}(T_{\varepsilon'})$ if $\varepsilon_1 = L$ and $T_\varepsilon = \text{shift}(T_{\varepsilon'}) \cup \{2a-1\}$ if $\varepsilon_1 = R$, where $\varepsilon' = (\varepsilon_2, \dots, \varepsilon_a)$ and $T_{\varepsilon'} \in S_{a-1}$.

By the inductive hypothesis on $\varepsilon' \in \{L, R\}^{a-1}$,

$$T_{\varepsilon'} = \{i-1 : \varepsilon'_i = L\} \cup \{2(a-1)-i : \varepsilon'_i = R\}.$$

Shifting by $+1$:

$$\text{shift}(T_{\varepsilon'}) = \{i : \varepsilon'_i = L\} \cup \{2a-1-i : \varepsilon'_i = R\}.$$

Substituting $j = i+1$ (so j ranges over $\{2, \dots, a\}$ and $\varepsilon'_i = \varepsilon_j$):

$$\text{shift}(T_{\varepsilon'}) = \{j-1 : \varepsilon_j = L, 2 \leq j \leq a\} \cup \{2a-j : \varepsilon_j = R, 2 \leq j \leq a\}.$$

If $\varepsilon_1 = L$, prepending $\{0\} = \{1-1 : \varepsilon_1 = L\}$ extends the union to $j \in \{1, \dots, a\}$, matching the claim. If $\varepsilon_1 = R$, appending $\{2a-1\} = \{2a-1 : \varepsilon_1 = R\}$ does the same. □

Remark 2. The closed form gives an elegant “ L/R -side” partition of $[2a]$: the L -bits contribute to the “low half” $\{0, \dots, a-1\}$ with the i -th L -bit at position $i-1$, and the R -bits contribute to the “high half” $\{a, \dots, 2a-1\}$ with the i -th R -bit at position $2a-i$ (mirror-symmetric).

3 Case alternation along the trajectory

We need to know, at each step k of the trajectory, whether the pair $\{P_k^+, P_k^-\}$ is in Case I (e_2 in column 1 of P_k^+ , e_3 in column 2) or Case II (e_2 in column 2, e_3 in column 1), in the May-17 nomenclature. The following determines the case sequence from τ .

Lemma 3 (Case at step k). *For each $k \in \{0, 1, \dots, a-1\}$, the pair $\{P_k^+, P_k^-\}$ at the start of step k is:*

- $k = 0$: Case II.
- $k \geq 1$: Case I if $\tau_k = L$; Case II if $\tau_k = R$.

Proof. Base $k = 0$. $P_0^+ = T_A(T')$ has $n - 1$ at column 2 row a and n at column 1 row $n - a$. So $e_2 = n - 1$ is in column 2 of P_0^+ and $e_3 = n$ is in column 1: Case II.

Inductive step $k \rightarrow k + 1$ (for $k \geq 0$). After the kill/expand at step k (May-17 Lemma 3.1), the new pair is $\{P_{k+1}^+, P_{k+1}^- = s_{e_1^{(k)}} P_{k+1}^+\}$ where $e_1^{(k)} = n - 2 - k$. The new $e_2^{(k+1)} = n - 1 - (k + 1) = n - 2 - k = e_1^{(k)}$. So the case at step $k + 1$ is determined by: is $e_1^{(k)} \in \sigma(P_{k+1}^+)$?

By the May-17 Corollary on encoding-bit identification (Cor. 5.5): $e_1^{(k)} = n - 2 - k \in \sigma(P_k^+)$ iff $\tau_{k+1} = R$. The transposition $s_{e_2^{(k)}}$ does not move $e_1^{(k)}$, so $e_1^{(k)} \in \sigma(P_k^-) \iff e_1^{(k)} \in \sigma(P_k^+)$. Hence in either survivor (whether $P_{k+1}^+ = P_k^+$ or $P_{k+1}^+ = P_k^-$),

$$e_1^{(k)} \in \sigma(P_{k+1}^+) \iff \tau_{k+1} = R.$$

The new $e_2^{(k+1)} = e_1^{(k)}$; “ $e_2^{(k+1)} \in \sigma$ ” means column 2, which is Case II. Hence Case II at step $k + 1$ iff $\tau_{k+1} = R$, i.e. Case I iff $\tau_{k+1} = L$. $\square$

Remark 4. The case is governed by the sub-case (a/b) of the *previous* step: sub-case (a) $\Rightarrow$ Case I next; sub-case (b) $\Rightarrow$ Case II next. Equivalently, sub-case (a) and Case I are both “column 1” phenomena, while sub-case (b) and Case II are both “column 2”.

4 Three-piece σ -decomposition

For each $k \in \{0, 1, \dots, a - 1\}$, partition

$$\{n - 2a + 1, \dots, n\} = \underbrace{\{n - 2a + 1, \dots, n - 2 - k\}}_{\text{lower}} \sqcup \underbrace{\{n - 1 - k, n - k\}}_{\text{middle}} \sqcup \underbrace{\{n - k + 1, \dots, n\}}_{\text{upper}}.$$

For $k = a - 1$ the three blocks have sizes $a - 1, 2, a - 1$. Set:

$$A_k := \sigma(P_k^+) \cap \{n - 2a + 1, \dots, n - 2 - k\}, \quad (1)$$

$$M_k := \sigma(P_k^+) \cap \{n - 1 - k, n - k\}, \quad (2)$$

$$U_k := \sigma(P_k^+) \cap \{n - k + 1, \dots, n\}. \quad (3)$$

Then $\sigma(P_k^+) = A_k \sqcup M_k \sqcup U_k$ (and similarly with P_k^- , with M_k swapped and A_k, U_k unchanged since $P_k^- = s_{n-1-k} P_k^+$ only moves entries in $\{n - 1 - k, n - k\}$).

Lemma 5 (Three-piece formula). *For each $k \in \{0, 1, \dots, a - 1\}$:*

- (i) $A_k = \sigma(T') \cap \{n - 2a + 1, \dots, n - 2 - k\}$. (*Lower part.*)
- (ii) $U_k = \{n - j : 0 \leq j \leq k - 1, \tau_{j+1} = R\}$. (*Upper part.*)
- (iii) $M_k = \{n - 1 - k\}$ if step k is Case II, and $\{n - k\}$ if Case I.

Proof. (i) This is the May-17 σ -invariant (Proposition 5.1).

(iii) Immediate from the definitions of Cases I, II in May-17 Lemma 3.1: at the start of step k the pair is a T_{n-1-k} -2-block, P_k^+ has $e_2 = n - 1 - k$ and $e_3 = n - k$ at different columns of λ , with one in column 2 (yielding the σ -membership) and the other in column 1.

(ii) Induction on k . Base $k = 0$: $U_0 = \sigma(P_0^+) \cap \{n + 1, \dots, n\} = \emptyset$, matching the empty product on the right.

Inductive step $k \rightarrow k+1$. $U_{k+1} \cap \{n-k+1, \dots, n\} = \sigma(P_{k+1}^+) \cap \{n-k+1, \dots, n\}$. Now P_{k+1}^+ is the survivor of $\{P_k^+, P_k^-\}$, both of which agree on entries *outside* $\{e_2^{(k)}, e_3^{(k)}\} = \{n-1-k, n-k\}$, in particular on $\{n-k+1, \dots, n\}$. So

$$\sigma(P_{k+1}^+) \cap \{n-k+1, \dots, n\} = \sigma(P_k^+) \cap \{n-k+1, \dots, n\} = U_k.$$

Then $U_{k+1} = U_k \cup (\{n-k\} \cap \sigma(P_{k+1}^+))$ (adding membership at the new “boundary” position $n-k$, which moves from middle at step k to upper at step $k+1$).

Whether $n-k \in \sigma(P_{k+1}^+)$. This depends on the sub-case at step k . By May-17 Lemma 3.1 case analysis:

- Case I.a: $P_{k+1}^+ = P_k^-$. In Case I, $e_3 = n-k$ is in column 2 of P_k^+ , hence in column 1 of $P_k^- = s_{e_2} P_k^+$. So $n-k \notin \sigma(P_{k+1}^+)$.
- Case I.b: $P_{k+1}^+ = P_k^+$, e_3 in column 2. So $n-k \in \sigma(P_{k+1}^+)$.
- Case II.a: $P_{k+1}^+ = P_k^+$. In Case II, e_3 in column 1. So $n-k \notin \sigma(P_{k+1}^+)$.
- Case II.b: $P_{k+1}^+ = P_k^-$, e_3 in column 2 of P_k^- . So $n-k \in \sigma(P_{k+1}^+)$.

In all four: $n-k \in \sigma(P_{k+1}^+)$ iff the sub-case is (b), iff (by May-17 Cor. 5.5) $\tau_{k+1} = R$.

Hence $U_{k+1} = U_k \cup (\{n-k\} \text{ if } \tau_{k+1} = R, \text{ else } \emptyset)$. Combined with the inductive hypothesis,

$$U_{k+1} = \{n-j : 0 \leq j \leq k, \tau_{j+1} = R\}. \quad \square$$

5 Final pair identification

Theorem 6 (Final pair = $\{T_{\tau L}, T_{\tau R}\}$). *For every $a \geq 2$, every $n \geq 3a$, every $T' \in S_{a-1}^{(n-2)}$ with encoding $\tau = \varepsilon(T') \in \{L, R\}^{a-1}$,*

$$\{\bar{\sigma}(P_{a-1}^+(T')), \bar{\sigma}(P_{a-1}^-(T'))\} = \{T_{\tau L}, T_{\tau R}\}.$$

In particular, $\bar{\sigma}(P_{a-1}^+), \bar{\sigma}(P_{a-1}^-) \in S_a$ with encoding prefix τ and last bit varying over $\{L, R\}$.

Proof. Apply Lemma 5 at $k = a-1$ and shift by $-(n-2a+1)$:

$$\begin{aligned} \bar{A}_{a-1} &= \bar{\sigma}(T') \cap \{0, 1, \dots, a-2\}, \\ \bar{M}_{a-1} &= \begin{cases} \{a-1\} & \text{Case II at step } a-1 \\ \{a\} & \text{Case I at step } a-1 \end{cases}, \\ \bar{U}_{a-1} &= \{2a-1-j : 0 \leq j \leq a-2, \tau_{j+1} = R\}. \end{aligned}$$

By Lemma 3, the case at step $a-1$ is: Case I iff $\tau_{a-1} = L$ (so $\bar{M}_{a-1} = \{a\}$ in that case), Case II iff $\tau_{a-1} = R$ (so $\bar{M}_{a-1} = \{a-1\}$).

Simplifying $\bar{A}_{a-1}$ and $\bar{U}_{a-1}$. By the closed-form Lemma 1 applied to $\bar{\sigma}(T') = T_\tau \in S_{a-1}$,

$$\bar{\sigma}(T') = \{i-1 : \tau_i = L\} \cup \{2(a-1)-i : \tau_i = R\} \quad (1 \leq i \leq a-1).$$

The L -contributions lie in $\{0, \dots, a-2\}$ and the R -contributions in $\{a-1, \dots, 2a-3\}$ (Lemma 1 Remark). Restricting to $\{0, \dots, a-2\}$:

$$\bar{A}_{a-1} = \{i-1 : \tau_i = L, 1 \leq i \leq a-1\}.$$

For $\bar{U}_{a-1}$, substitute $i = j+1$:

$$\bar{U}_{a-1} = \{2a-i : \tau_i = R, 1 \leq i \leq a-1\}.$$

Comparison with the closed form for S_a . By Lemma 1 applied to $\varepsilon = \tau \cdot b \in \{L, R\}^a$ where $b \in \{L, R\}$ is the final bit:

$$\begin{aligned} T_{\tau L} &= \{i-1 : \tau_i = L, 1 \leq i \leq a-1\} \cup \{a-1\} \cup \{2a-i : \tau_i = R, 1 \leq i \leq a-1\}, \\ T_{\tau R} &= \{i-1 : \tau_i = L, 1 \leq i \leq a-1\} \cup \{a\} \cup \{2a-i : \tau_i = R, 1 \leq i \leq a-1\} \end{aligned}$$

(the bit at position a contributes $a-1$ if L , a if R). Equivalently,

$$T_{\tau L} = \bar{A}_{a-1} \cup \{a-1\} \cup \bar{U}_{a-1}, \quad T_{\tau R} = \bar{A}_{a-1} \cup \{a\} \cup \bar{U}_{a-1}.$$

Two cases.

- $\tau_{a-1} = L$: Case I at step $a-1$. $\bar{\sigma}(P_{a-1}^+) = \bar{A}_{a-1} \cup \{a\} \cup \bar{U}_{a-1} = T_{\tau R}$. And $\bar{\sigma}(P_{a-1}^-) = \bar{A}_{a-1} \cup \{a-1\} \cup \bar{U}_{a-1} = T_{\tau L}$ (swap the $\bar{M}$ piece, $\bar{A}$ and $\bar{U}$ unchanged).
- $\tau_{a-1} = R$: Case II at step $a-1$. $\bar{\sigma}(P_{a-1}^+) = T_{\tau L}$, $\bar{\sigma}(P_{a-1}^-) = T_{\tau R}$.

Either way, $\{\bar{\sigma}(P_{a-1}^+), \bar{\sigma}(P_{a-1}^-)\} = \{T_{\tau L}, T_{\tau R}\}$. $\square$

Remark 7 (Which-is-which depends on the path). The assignment $P_{a-1}^+ \leftrightarrow T_{\tau L}$ vs. $T_{\tau R}$ depends on τ_{a-1} . At $a=2$ with $\tau = L$ (so $\tau_{a-1} = L$): $P_1^+ \leftrightarrow T_{LR}$ (verified in May-17 §5.2 by hand). At $a=2, \tau = R$: $P_1^+ \leftrightarrow T_{RL}$. The asymmetry comes from the case-flip rule: $\tau_{a-1} = L$ forces Case I at the last step, where P^+ ends with $\bar{M} = \{a\}$ (matching $T_{\tau R}$).

6 Consequences

Theorem 8 (Support = $S_a^{(n)}$ unconditionally). *For every $a \geq 2$ and $n \geq 3a$, $B_{n-a+1}^{(\lambda)} V_\lambda$ is exactly 1-dimensional with seminormal support $S_a^{(n)}$ (of size $|S_a^{(n)}| = 2^a$), all coefficients in $\mathbb{Q}(q)$ nonzero.*

Proof. The dimension bound $\dim B_{n-a+1}^{(\lambda)} V_\lambda \leq 1$ was established in May-17 §4.2. By May-15-gen Theorem 3.2,

$$B_{n-a+1}^{(\lambda)} V_\lambda = \mathbb{Q}(q) \cdot \sum_{T' \in \text{supp}(u^{(\mu_1)})} c_{T'} w_{a-1}(T'),$$

where $u^{(\mu_1)}$ generates $B_{n-a}^{(\mu_1)} V_{\mu_1}$, which by induction at $a-1$ has support exactly $S_{a-1}^{(n-2)}$ and is 1-dimensional, so $|\text{supp}(u^{(\mu_1)})| = 2^{a-1}$ with nonzero coefficients $c_{T'}$.

By Theorem 6, $w_{a-1}(T')$ has support $\{T_{\tau L}, T_{\tau R}\}$ where $\tau = \varepsilon(T')$. As T' ranges over $S_{a-1}^{(n-2)}$, τ ranges over $\{L, R\}^{a-1}$ bijectively, hence

$$\bigsqcup_{T' \in S_{a-1}^{(n-2)}} \{T_{\tau(T')L}, T_{\tau(T')R}\} = S_a^{(n)},$$

because $\varepsilon : S_a \rightarrow \{L, R\}^a$ is a bijection and the pairs $\{T_{\tau L}, T_{\tau R}\}$ partition S_a according to their first $a-1$ encoding bits. The supports are pairwise disjoint (different τ 's give different prefixes, hence different pairs). Therefore $\bigcup_{T'} \text{supp}(w_{a-1}(T')) = S_a^{(n)}$.

Nonzero coefficients. By May-17 (P4) at $k = a-1$, both seminormal coefficients of $w_{a-1}(T')$ are nonzero in $\mathbb{Q}(q)$. By the bijection $T' \leftrightarrow \tau$ and the pair-partition, the linear combination $\sum_{T'} c_{T'} w_{a-1}(T')$ has coefficient at each $T_\varepsilon \in S_a$ equal to a single (nonzero) contribution from the unique T' with $\varepsilon(T')$ -matching prefix. Hence the support is exactly $S_a^{(n)}$ with all $\mathbb{Q}(q)$ -coefficients nonzero, and the sum is nonzero, confirming dimension 1. $\square$

Corollary 9 (Rank-zero theorem at $(2^a, 1^{n-2a})$, unconditional on conjectural support). *For every $a \geq 1$ and $n \geq 3a$, $\Pi^{S_n}|_{V_{(2^a, 1^{n-2a})}} = 0$, conditional only on the Phase A endpoint $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$ at general a .*

Proof. May-17 Theorem 4.1 gives this conditionally on Conjecture 5.6. Theorem 8 above closes that conjecture. $\square$

7 Computational verification

We verified the final pair identification by direct σ -level trajectory simulation: for each $T' \in S_{a-1}^{(n-2)}$, simulate the kill/expand trajectory using only the structural rule (kill iff e_1 and e_2 in the same column), and compare the final pair against the predicted $\{T_{\tau L}, T_{\tau R}\}$ from the closed form.

a	n	$ S_{a-1}^{(n-2)} $	result
2	6, 7, 8	2 each	PASS (all 2 trajectories match)
3	9, 10, 11	4 each	PASS (all 4 trajectories match)
4	12, 13, 14	8 each	PASS (all 8 trajectories match)
5	15, 16	16 each	PASS (all 16 trajectories match)
6	18	32	PASS (all 32 trajectories match)

Script: `~/projects/scratch/2026-05-18-final-pair/verify_trajectory.py`. Total of 12 cases at $a \in \{2, 3, 4, 5, 6\}$, covering $2 + 2 + 2 + 4 + 4 + 4 + 8 + 8 + 8 + 16 + 16 + 32 = 106$ trajectory simulations. All match the closed-form prediction $\{T_{\tau L}, T_{\tau R}\}$.

A separate script (`verify_closed_form.py`) verified the closed-form S_a formula against the recursive definition for $a \leq 6$ (all $|S_a| = 2^a$ elements, with encoding round-trips through both directions).

We also performed hand verifications at $a = 3$ in §8.

8 Examples

8.1 $a = 3, n = 9, \tau = LL$

T' with $\bar{\sigma}(T') = \{0, 1\}$, i.e. $\sigma(T') = \{4, 5\}$. The trajectory:

1. $P_0^+ = T_A, \sigma = \{4, 5, 8\}$. Step 0 ($e_1 = 7$): $7 \notin \sigma(T')$, sub-case (a), Case II.a. Survivor $P_1^+ = T_A$.
2. $P_1^+ = T_A$. Step 1 ($e_1 = 6$): $6 \notin \sigma(T_A)$, sub-case (a), Case I.a (Case I per Lemma 3 since $\tau_1 = L$). Survivor $P_2^+ = P_1^- = s_7 T_A, \sigma = \{4, 5, 7\}$.
3. Final pair: $\{s_7 T_A, s_6 s_7 T_A\}$, σ -values $\{\{4, 5, 7\}, \{4, 5, 6\}\}$. Shifted by -4 : $\{\{0, 1, 3\}, \{0, 1, 2\}\} = \{T_{LLR}, T_{LLL}\} = \{T_{\tau R}, T_{\tau L}\}$. $\checkmark$

8.2 $a = 3, n = 9, \tau = LR$

T' with $\bar{\sigma}(T') = \{0, 2\}$, i.e. $\sigma(T') = \{4, 6\}$.

1. $P_0^+ = T_A, \sigma = \{4, 6, 8\}$. Step 0 ($e_1 = 7$): $7 \notin \sigma(T')$, Case II.a. Survivor $P_1^+ = T_A$.
2. $P_1^+ = T_A$. Step 1 ($e_1 = 6$): $6 \in \sigma(T_A)$, sub-case (b), Case I.b. Survivor $P_2^+ = P_1^+ = T_A, \sigma = \{4, 6, 8\}$.

3. Final pair: $\{T_A, s_6 T_A\}$, σ -values $\{\{4, 6, 8\}, \{4, 7, 8\}\}$. Shifted: $\{\{0, 2, 4\}, \{0, 3, 4\}\} = \{T_{LRL}, T_{LRR}\} = \{T_{\tau L}, T_{\tau R}\}$. ✓

8.3 Formula check

By Theorem 6, with $\tau = LR$:

$$\begin{aligned}\bar{A}_{a-1} &= \{i - 1 : \tau_i = L, 1 \leq i \leq 2\} = \{0\}, \\ \bar{U}_{a-1} &= \{2a - i : \tau_i = R, 1 \leq i \leq 2\} = \{4\}, \\ \bar{M}_{a-1}(P^+) &= \{a - 1\} = \{2\} \quad (\text{Case II, since } \tau_{a-1} = \tau_2 = R).\end{aligned}$$

Predicting $\bar{\sigma}(P_2^+) = \{0, 2, 4\} = T_{\tau L} = T_{LRL}$. ✓

9 Honest assessment

What this paper rigorously proves.

1. Closed form (Lemma 1) for elements of S_a in terms of their encoding. Elementary induction.
2. Case alternation at each step (Lemma 3): each step's case (I vs. II) is determined by the corresponding encoding bit, with step 0 always Case II. Follows from May-17 Cor. 5.5.
3. Three-piece σ -decomposition (Lemma 5): $\sigma(P_k^+)$ splits into lower (May-17 invariant), middle (case), upper (τ -tracking).
4. Final pair identification (Theorem 6): $\{\bar{\sigma}(P_{a-1}^\pm)\} = \{T_{\tau L}, T_{\tau R}\}$.
5. Support identification (Theorem 8): the support of $B_{n-a+1}^{(\lambda)} V_\lambda$ is exactly $S_a^{(n)}$, $|S_a^{(n)}| = 2^a$, dimension 1, all coefficients nonzero.
6. Rank-zero theorem (Corollary 9) for the family $\lambda = (2^a, 1^{n-2a})$ at every $a \geq 1$, $n \geq 3a$. Conditional on Phase A endpoint.

What remains conditional.

1. Phase A endpoint at general $a \geq 4$: the assertion $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$. Proved at $a = 2$ (May-13 Prop. 3.1) and $a = 3$ (May-14 Prop. 3.1); asserted mechanical for $a \geq 4$ in May-15-gen Theorem 2.1. Computationally verified for $a = 4, 5$ at small n . This is an orthogonal piece of work, structurally independent of the per-trajectory framework.
2. Generic non-vanishing of the summed image $\sum_{T'} c_{T'} w_{a-1}(T')$: handled in Theorem 8 via the disjoint-pair argument and (P4) of the May-17 invariant.

What's NOT conditional anymore. The support-size 2^a conjecture and the L/R encoding identification — the explicit combinatorial content of the May-13/May-15/May-16 case enumerations — is now proved *uniformly* via:

- The May-17 four-case structural lemma (per-trajectory invariant).
- The May-17 σ -invariant (lower-range freezing).
- The closed form for S_a (Lemma 1).
- The three-piece σ -decomposition (Lemma 5).

- Theorem 6.

No case-by-case enumeration; no encoding-bit bookkeeping beyond the clean “ $\bar{M}_{a-1} \in \{\{a-1\}, \{a\}\}$ ” final-step distinction.

Big picture. The May-13/15/16/17 program has been moving toward exactly this clean statement. May-13 enumerated by hand at $a = 2$. May-15-a3 enumerated by hand at $a = 3$. May-16 identified the per-trajectory framework and conjectured the uniform support flow. May-17 closed the uniform per-trajectory invariant and the σ -invariant, isolating the residue. This paper closes the residue via the closed form for S_a and the upper- U_k tracking.

The proof is short because the underlying object is well-structured: elements of S_a have a clean closed form, and $\sigma(P_k^+)$ decomposes into three independent pieces that each have a clean description in terms of τ . The original case enumeration was illusorily complex; once the right invariants are isolated, the identification is mechanical.